# Pratik Wadekar

24M0843

**Computer Science & Engineering** — **M.Tech**

Indian Institute of Technology, Bombay — Male

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| Examination | Board/University | Institute | Year | CPI/% |
|---|---|---|---|---|
| Post Graduation | IIT Bombay | IIT Bombay | 2024-27 | 8.76 |
| Graduation | Shivaji University, Kolhapur | RIT Rajaramnagar | 2019-23 | 8.98 |
| Intermediate/+2 | Maharashtra State Board | The New College, Kolhapur | 2017-19 | 85.85% |
| Matriculation | Maharashtra State Board | New English School Bahireshwar | 2016-17 | 95.80 |

## COURSE PROJECTS

- **CRISP Prefetching for Limit Order Book Workloads**, *(C++, ChampSim, Intel PIN, Bash)* *(Oct'25-Nov'25)*
  - Profiled and optimized a prefetcher within the ChampSim simulator using Intel VTune to implement CRISP logic.
  - Engineered delinquent-load detection, backward-slice construction, and a critical-PC table to prioritize latency-critical instructions.
  - Built an Intel PIN pipeline to generate ChampSim-compatible LOB traces, evaluating across 5 workloads via 25M-instruction simulations to achieve up to a 4.98% IPC gain.
  - Released ChampSim patches, PIN tracers, and Bash automation scripts to ensure reproducible experimental results.
- **Conductor Container**, *(Bash)* *(Mar'25-Apr'25)*
  - Architected a custom container runtime leveraging Linux namespaces and `chroot`, fully supporting `run`, `ps`, `stop`, `exec`, and `overlayfs` layering.
  - Parsed `Conductorfile` configurations (FROM, COPY, RUN) for custom image builds and implemented isolated networking pipelines using `veth` pairs and namespace routing.
  - Developed an orchestrator script to deploy and bridge two microservices in isolated containers with port forwarding.
- **DeFi DEX Platform**, *(Solidity, JS, OpenZeppelin)* *(Apr'25)*
  - Deployed ERC-20 tokens on the Remix VM and implemented a constant-product Automated Market Maker (AMM) with a 0.3% swap fee, dynamic pricing, and LP token mint/burn mechanics.
  - Developed JavaScript scripts to simulate deposits, withdrawals, and swaps while actively tracking TVL, volume, fees, and slippage.
  - Engineered an automated arbitrage smart contract to detect and execute multi-hop trades, ensuring protocol security via strict input validation and safe math operations.
- **Kernel in Rust**, *(Rust, x86_64, QEMU)* *(Oct'24-Nov'24)*
  - Developed a 64-bit bare-metal (`no_std`) OS kernel from scratch, implementing a virtual memory manager with paging and a fixed-size block allocator for the kernel heap.
  - Engineered low-level CPU exception handlers, managed hardware interrupts via an 8259 PIC, and configured the GDT/TSS to ensure stable stack switching.
  - Built a thread-safe VGA text driver and an interactive shell interpreting native PS/2 keyboard scancodes for terminal command execution.
  - Architected a hierarchical in-memory virtual file system supporting standard directory traversal and CRUD operations.
  - Integrated an automated kernel-level testing framework utilizing QEMU's `isa-debug-exit` mechanism for synchronous assertions.
- **Placement Portal**, *(Django REST Framework)* *(Nov'24)*
  - Designed and developed the core RESTful APIs for a comprehensive Placement Portal using Django and Django REST Framework (DRF).
  - Implemented a secure, custom role-based authentication system to manage varied permissions for students, administrators, and recruiters.
  - Designed relational database schemas (Users, Job Posts, Applications, interactive Feeds, and Comments) ensuring data integrity and efficient querying.
  - Engineered a complete job application flow, enabling companies to publish job descriptions and students to seamlessly apply and track their statuses.

- **KV Caching - nanoGPT Inference Optimization**, *(PyTorch, Transformers)* *(Feb'26)*
  - Implemented Key-Value (KV) caching in the nanoGPT transformer architecture to optimize autoregressive inference.
  - Modified the self-attention pipeline to reuse cached keys and values across decoding steps, avoiding redundant computation.
  - Improved token generation latency and overall inference efficiency during text generation.
- **GPU-Optimized Matrix Multiplication**, *(CUDA, C++)* *(March'26)*
  - Implemented multiple CUDA kernel strategies for matrix multiplication, exploring different thread-to-data mappings and tiling approaches.
  - Optimized GPU memory access patterns by using strided and contiguous loads to improve thread utilization and memory coalescing.

## TECHNICAL SKILLS

- **Languages**: C++, C, Rust, Python, Java, Shell, CUDA
- **Frameworks & ML**: Django, PyTorch, Transformers, Self-Attention, Deep Learning
- **DevOps & Cloud**: Docker, Kubernetes, Ansible, CI/CD Pipelines, AWS DevOps, CloudFormation, IAM, Fargate, Lambda, EC2
- **Tools & Databases**: Git, Linux, Neovim, PostgreSQL, SQLite, LDAP, Mail Server, AWS, GCP

## WORK EXPERIENCE & INTERNSHIPS

- **System Administrator (RA)** | ***IIT Bombay*** *(Aug'24-Present)*
  - Automated laboratory environment setup and software deployments across multiple systems using Ansible, reducing manual configuration overhead.
  - Managing and maintaining critical CSE infrastructure services including Mail Server, LDAP authentication, and Roundcube webmail.
  - Administered Linux-based systems, ensuring reliable service operation, user account management, and secure access control across lab environments.
- **AWS DevOps Engineer** | ***Ignitiv Technologies Pvt. Ltd*** *(Jul'23-Jul'24)*
  - Managed AWS accounts within an organization-wide environment, provisioning and maintaining cloud infrastructure including EC2 instances and CloudFront distributions.
  - Automated application deployments by designing CI/CD pipelines using Bitbucket Pipelines, improving deployment reliability and reducing manual release effort.
  - Designed and implemented a serverless data ingestion workflow for aggregating order data from multiple e-commerce platforms: scheduled events triggered AWS Lambda functions that launched AWS Fargate tasks to run containerized jobs for fetching and transferring data to KIBO.
  - Administered AWS IAM policies and role-based access controls, ensuring secure and organized access management across teams.
- **AWS Developer Intern** | ***KONE Elevator & Escalators*** *(Jan'23–Jun'23)*
  - Designed and implemented an alarms and event-trigger notification pipeline for monitoring system events across AWS infrastructure.
  - Developed serverless applications using Python and the AWS SAM framework, integrating services including Kinesis, Lambda, DynamoDB, API Gateway, SQS, CloudWatch, Step Functions, Secrets Manager, and SSM.
  - Built a disaster-handling pipeline to detect and notify teams when critical AWS resources were deleted from source accounts using CloudTrail, Lambda, and SNS.
  - Maintained source control and CI workflows using GitLab, enabling structured development and deployment of serverless infrastructure.
  - Implemented IAM-based access control and secure secret management to ensure safe interaction between AWS services.

## ACHIEVEMENTS & ACADEMICS

- **Competitive Programming**: Leetcode (700+ problems, 3 Star), Codeforces (250+ problems).
- **GATE 2024**: Qualified in Computer Science Engineering (CSE) with an All India Rank (AIR) of 1532 out of 123967.